

Alee Hafeez

Data Solutions Architect | Senior Data Engineer | Big Data Engineer
Syracuse, NY 13217 | +1(845)207-6832 | alee.farooq1337@gmail.com

Profile

With 8+ years of experience designing scalable data platforms and high-throughput pipelines across AWS, Azure, and GCP. Expertise in building batch and real-time data processing systems using Spark, Kafka, Airflow, dbt, and modern cloud data warehouses such as Snowflake, BigQuery, and Redshift. Strong experience in data modeling, distributed processing, and event-driven architectures, with hands-on expertise in Python, SQL, and Infrastructure as Code using Terraform to build reliable, production-grade data systems.

Proven track record of leading cloud migrations, architecting distributed data systems, and enabling analytics and machine learning platforms at scale. Skilled in designing reliable ETL/ELT frameworks, implementing data governance and quality standards, and building infrastructure-as-code deployments using Terraform, Docker, and Kubernetes, with expertise in Data Modeling, CI/CD, and Data Warehousing.

Collaborates closely with data scientists, analysts, and product teams to deliver trusted, scalable, and production-grade data solutions that support advanced analytics, operational reporting, and machine learning initiatives.

Skills

- **Data Engineering and Pipelines:** Apache Airflow, dbt, AWS Glue, Azure Data Factory, Luigi, Prefect, Dagster, Scalable ETL/ELT pipelines, data quality frameworks (Great Expectations, Deequ), DAG orchestration, data validation, metadata management, backfilling and recovery strategies
- **Big Data and Distributed Processing:** Apache Spark, Apache Flink, Databricks, Hadoop, Amazon EMR, Google Dataflow, Dask, Ray, Apache Beam, Delta Lake, Iceberg, Hudi
- **Programming Languages:** Python (pandas, NumPy, PySpark, fastapi, typer), SQL (T-SQL, PL/pgSQL), Scala, Bash, YAML, JSON, Java (basic for Spark), Go
- **Machine Learning:** Model training and evaluation, ML pipeline orchestration (Airflow, Kubeflow, Vertex AI Pipelines), feature engineering, model versioning (MLflow, DVC), ML deployment (Docker, Kubernetes, REST APIs for inference, model monitoring with Prometheus and custom metrics), batch and real-time inference
- **Streaming and Messaging Systems:** Apache Kafka, Amazon Kinesis, Google Pub/Sub, Spark Streaming, Apache Pulsar, Kafka Streams, Flink CEP, Change Data Capture (Debezium, Kafka Connect)
- **Cloud Platforms:** AWS (S3, Redshift, Lambda, Glue, Athena, RDS, EC2, CloudFormation), GCP (BigQuery, Dataflow, Pub/Sub, Vertex AI, Cloud Composer), Azure (Synapse, Data Lake, Data Factory, Event Hubs)
- **Data Warehousing and Analytics:** Snowflake, BigQuery, Redshift, Azure Synapse, dbt, Looker, Mode Analytics, Tableau, Power BI, Apache Superset, Sigma Computing
- **DevOps and Infrastructure as Code (IaC):** Terraform, CloudFormation, Pulumi, Docker, Kubernetes, Helm, Jenkins, GitHub Actions, GitLab CI/CD, ArgoCD, CircleCI
- **Monitoring and Observability:** Prometheus, Grafana, Datadog, CloudWatch, OpenTelemetry, ELK stack (Elasticsearch, Logstash, Kibana), Splunk, New Relic, custom logging and alerting with Fluentd and Loki
- **Security, Governance and Compliance:** Data cataloging and lineage (Apache Atlas, DataHub, Amundsen, Collibra, Alation), RBAC/ABAC, encryption (at rest/in transit), secrets management (Vault, AWS Secrets Manager, Azure Key Vault), GDPR, HIPAA, SOC 2
- **Database Systems:** PostgreSQL, MySQL, Oracle, MongoDB, Cassandra, DynamoDB, Redis, ClickHouse, TimescaleDB, Elasticsearch
- **Data Modeling and Architecture:** Dimensional modeling, Data Vault, star/snowflake schema design, normalized/denormalized data models, event-driven architectures, lambda and kappa architecture, CDC patterns, schema evolution and governance

Experience

Lead Data Engineer, Gold Coast Health Plan

Oct 2022 – Present

- Led the architecture and deployment of a multi-cloud data platform across AWS, GCP, and Azure, supporting 30+ production pipelines and processing 5TB+ of healthcare data daily
- Designed and implemented end-to-end ETL/ELT pipelines using Airflow, dbt, and Apache Beam, reducing data processing latency by 40 percent and improving data availability for analytics teams.
- Built real-time streaming pipelines using Apache Kafka, Amazon Kinesis, and Flink, enabling sub-second ingestion of 2M+ events per hour for operational dashboards and monitoring systems.
- Optimized warehouse workloads in BigQuery, Snowflake, and Redshift, improving complex analytics query performance by 35–50 percent
- Implemented MLOps pipelines using Kubeflow, MLflow, Docker, and Kubernetes, reducing model deployment time from weeks to under 24 hours.
- Automated infrastructure provisioning with Terraform and Helm, reducing manual cloud deployment effort by 60 percent and improving environment consistency across teams.
- Established enterprise data governance and lineage frameworks using DataHub and Great Expectations, increasing pipeline reliability and reducing data quality incidents by 30 percent.
- Mentoring engineering teams and collaborating with product, analytics, and ML teams to deliver resilient, scalable, and cost-effective data solutions in a fast-paced environment.
- Mentored 5+ data engineers and established engineering best practices for pipeline design, monitoring, and CI/CD.

Senior Data Engineer, Analytics8

Jun 2019 – Sep 2022

- Architected and maintained scalable ETL pipelines using Airflow and AWS Glue to process and normalize advertising data across multiple sources.
- Led migration of legacy data systems to a cloud-native stack on AWS (S3, Redshift, Lambda), improving performance and reducing operational overhead.
- Collaborated with cross-functional teams to design data models and warehouse schemas in Redshift and Snowflake, enabling faster analytics and reporting.
- Integrated Kafka for real-time data streaming and built robust ingestion pipelines with Python and Spark for time-sensitive ad delivery insights.
- Implemented CI/CD workflows using GitHub Actions and Terraform to automate deployments and infrastructure provisioning.
- Ensured data quality and observability through automated testing, monitoring (Datadog, CloudWatch), and lineage tracking using DataHub.

Data Engineer, Etleap

Apr 2017 – May 2019

- Joined an early-stage startup team where I worked closely with engineers to support the design and implementation of ETL pipelines, helping move data from various sources into centralized systems for analysis
- Wrote and maintained Python and SQL scripts to perform data cleaning, transformation, and enrichment tasks that improved the quality and usability of data for downstream reporting
- Gained hands-on experience with orchestration tools like Apache Airflow and AWS Glue by setting up simple DAGs and understanding how to schedule, monitor, and rerun jobs when needed
- Assisted in identifying and resolving pipeline failures by checking logs, validating data outputs, and shadowing senior engineers to better understand common reliability issues
- Worked within cloud-based environments using services like Amazon S3 for storage and Redshift for querying structured data, learning how cloud infrastructure supports scalable analytics
- Contributed to internal documentation and knowledge sharing, gaining insight into how startups operate cross-functionally and how to adapt quickly in a fast-paced, evolving environment

Education

Bachelors in Computer Science

Feb 2013 – Mar 2017